Software Engineering
UNIT-1

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Unit -1 Syllabus

Module /Unit		Course Modules/Contents	Hours	Marks Weightage
1		Introduction to Software Engineering	3	10%
	1.1	The story of Software development and issues faced		
	1.2	Need for Systematic process for addressing issues		
	1.3	Products, custom solutions, services, domains, Technologies		

What is a software?

- **Software** is a collection of executable programming code, associated libraries and documentations.
- Software, when made for a specific requirement is called **software product.**

- **Engineering** is developing products, using well-defined, scientific principles and methods.
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Software Development Activities

IEEE Definition of Software Engineering

- *The application of a systematic,disciplined,quantifiable approach to the development,operation and maintenance of software; that is, the application of engineering to software.*

Microsoft- July 2024 global outage

- A software update from CrowdStrike, an independent cybersecurity company, caused a global outage that impacted Microsoft's services. The outage began when the update caused a "blue screen of death" to appear on user's screens instead of the Windows OS booting up. The outage affected Windows-based desktop and laptops globally, including in India, the United States, and New Zealand. The outage caused ground operations at airports in India to be affected, and the Indian Computer Emergency Response Team (CERT-In) rated the incident as "Critical". Microsoft and CrowdStrike worked together to resolve the issues, and Microsoft posted instructions to remedy the situation on Windows endpoints.

Products in Software Engineering

- System Software: This includes operating systems and utility programs that manage hardware resources.
- Application Software: These are end-user applications designed for specific tasks, such as word processors or database management systems.
- Embedded Software: Software that is integrated into hardware devices, such as firmware in appliances or automotive systems.
- Web Applications: These applications run on web servers and are accessed via web browsers, covering a wide range of functionalities from e-commerce to social

2. Custom Solutions

- Tailored Development: Custom software development involves creating solutions from scratch to address unique business requirements.
- Integration Services: Custom solutions often require integrating existing systems with new software to ensure seamless operation across platforms.

Services in Software Engineering

- Consultation Services: Offering expertise in software architecture, design patterns, and technology selection.
- Development Services: Full-cycle software development services that cover planning, designing, coding, testing, and deployment.
- Maintenance and Support: Ongoing support for existing systems to ensure they remain functional and relevant over time.
- Quality Assurance Services: Testing services aimed at ensuring software reliability

Domains of Application

- Business Applications: Software designed for enterprise resource planning (ERP), customer relationship management (CRM), and other business processes.
- Scientific Computing: Applications that support scientific research and simulations in fields like physics, biology, and engineering.
- Safety-Critical Systems: Systems where failure could result in significant harm or loss, such as in healthcare devices or aerospace applications.
- Gaming and Entertainment: Software for video games and interactive media that requires high-performance graphics and real-time processing.

Technologies in Software Engineering

- Agile Development Methodologies: Emphasizing iterative development and flexibility to adapt to changing requirements.
- Modeling Languages (e.g., UML): Used for visualizing system architecture and design through diagrams that represent various aspects of the system.
- DevOps Practices: Integrating development and operations to improve collaboration, automate processes, and enhance deployment efficiency.
- Cloud Computing Technologies: Leveraging cloud infrastructure for scalable applications that can handle varying loads without significant upfront investment.
- Artificial Intelligence and Machine Learning: Incorporating AI/ML techniques into applications for enhanced functionality such as predictive analytics or natural language processing.